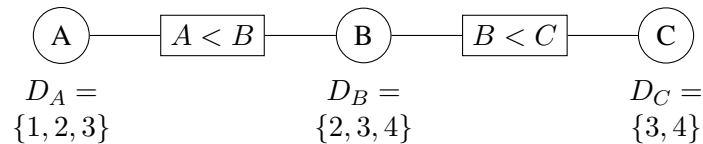


Exercise 1 :

- Which arcs in the above constraint network are arc consistent?
- How can the whole network be made arc consistent, i.e., which changes should be made to the domains of the variables making the network arc-consistent without eliminating potential solutions?

Solution:

- The arc $\langle B, B < C \rangle$ is not consistent, because for $B = 4$ there is no value in D_C that satisfies the constraint. All other arcs are arc consistent.
- First the value 4 has to be removed from D_B . Now the arc $\langle A, A < B \rangle$ no longer is consistent, because for $A = 3$ now no suitable value for B exists. After also removing 3 from D_A the network is arc consistent.

Exercise 2 :

Consider the following mini-Sudoku:

		4	
2			
		1	
3			

The empty fields have to be filled with numbers 1,2,3, or 4, such that each row, each column, and each of the 2×2 sub-squares contain each of these numbers exactly once.

- Formalize this problem as a constraint satisfaction problem: define an appropriate set of variables, and a set of constraints that define the solutions of the sudoku (hint: instead of using constraints as on the lecture slide, define a smaller set of constraints, each constraint expressing the full condition for one row, one column, or one sub-square. We have not discussed off-the-shelf formal languages for expressing such constraints, so you should try to express them as formally as possible).
- Draw the constraint network for this problem. If this gets too large, draw only the part of the network that is sufficient to answer the next question.
- Apply the generalized arc consistency algorithm to show that the top left square must contain a 1. Does the operation of the algorithm resemble how you would come to that conclusion yourself?

Solution:

Label the empty fields with variables like this:

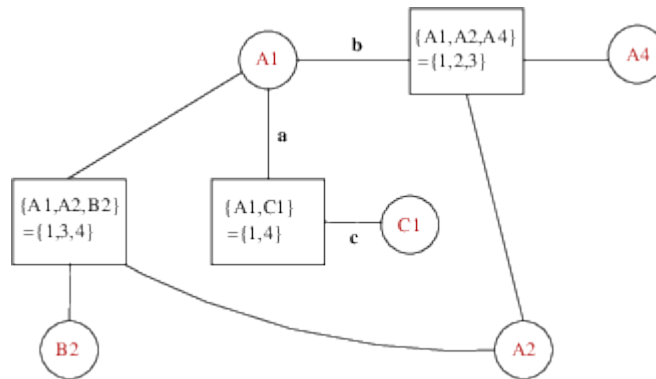
A1	A2	4	A4
2	B2	B3	B4
C1	C2	1	C4
3	D2	D3	D4

We now know that $A1, A2, A4$ combined must contain the values 1,2,3. We write this as a constraint in the form:

$$\{A1, A2, A4\} = \{1, 2, 3\}$$

Similarly, we define one constraint for each remaining row, column, and 2×2 sub-square.

A part of the constraint network now is:



Initially, we have $D_{A1} = D_{A2} = \dots D_{D4} = \{1, 2, 3, 4\}$. We first find that the arc **a** is not arc consistent, because for the values $A1 = 2$ or $A1 = 3$ the constraint $\{A1, C1\} = \{1, 4\}$ can not be satisfied. Thus, we delete 2 and 3 from D_{A1} . Since $D_{C1} = \{1, 2, 3, 4\}$ the arc now is arc consistent. Next inspect arc **b**. For $A1 = 4$ there are no values of $A2$ and $A4$ that could make the constraint satisfied, so 4 also is removed from D_{A1} . With $D_{A2} = D_{A4} = \{1, 2, 3, 4\}$ arc **b** now is arc consistent. At this point we have $D_{A1} = \{1\}$, i.e. we know that in any possible solution of the sudoku $A1$ must have value 1. We can now continue. For example, making arc **c** arc consistent immediately leads to the reduction $D_{C1} = \{4\}$.

Exercise 3 :

Consider the following constraint network $\gamma = (V, D, C)$:

- Variables: $V = \{a, b, c, d\}$.
- Domains: For all $v \in V$: $D_v = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$.

- Constraints: $2|a - c| > 3$; $b^2 - 3d < 9$; $b + 3 < c$.

Run the Generalized Arc Consistency(γ) algorithm, as specified in the lecture. Precisely, for each iteration of the while-loop, give the content of To-do-arcs at the start of the iteration, give the pair (x, c) removed from To-do-arcs, give the domain D_x of x after the call to $\text{Revise}(\gamma, x, c)$, and give the pairs (y, c') added into M .

Note: Initialize To-do-arcs as a lexicographically ordered list (i.e., (a, b) would be before (a, c) , both before (b, a) etc., if any of those exist). Furthermore, use to-do-arcs as a FIFO queue, i.e., always remove the element at the front and add new elements at the back.

Solution: Here is the required information for the iterations of the while-loop as executed. In this case, we always removed the element at the front first, and added new elements to the end.

For brevity, we call To-do-arcs M and use (a, c) to refer to variable a and the constraint C_{ac} .

1. M content: $\{(a, c), (b, c), (b, d), (c, a), (c, b), (d, b)\}$; pair selected: (a, c) ; No change, so nothing added to M .
2. M content: $\{(b, c), (b, d), (c, a), (c, b), (d, b)\}$; pair selected: (b, c) ; $D_b = \{1, 2, 3, 4, 5, 6\}$. Candidate (d, b) already present in M .
3. M content: $\{(b, d), (c, a), (c, b), (d, b)\}$; pair selected: (b, d) ; No change, so nothing added to M .
4. M content: $\{(c, a), (c, b), (d, b)\}$; pair selected: (c, a) ; No change, so nothing added to M .
5. M content: $\{(c, b), (d, b)\}$; pair selected: (c, b) ; $D_c = \{5, 6, 7, 8, 9, 10\}$. Candidate (a, c) added into M .
6. M content: $\{(d, b), (a, c)\}$; pair selected: (d, b) ; No change, so nothing added to M .
7. M content: $\{(a, c)\}$; pair selected: (a, c) ; No change, so nothing added to M .
8. M empty; return modified γ with reduced domains.

Exercise 4 :

Solve exercise 4.14 in PM about variable elimination.

Solution:

- r_1 and r_2 are removed. $r_{11}(B, C)$ is added.
- R_3, r_4 , and r_{11} are removed. $R_{12}(C, D, E)$ is added.

Exercise 5 :

Use Variable Elimination to solve the following CSP given by extensional constraints on Boolean variables A, B, C :

A	B	A	C	B	C
t	f	t	f	t	f
t	t	f	t	f	t
f	t				

Solution:

We eliminate B first. Joining the two tables containing B gives

A	B	B	C	A	B	C
t	f	t	f	t	f	t
t	t	f	t	t	t	f
f	t			f	t	f

Projecting on A, C then gives

A	C
t	t
t	f
f	f

We eliminate C next (the order in which we perform the elimination is not very important in this example). We now have two tables containing C , which we join:

A	C	A	C
t	f	t	t
f	t	t	f
		f	f

The result of this join now is the only constraint left. Therefore, the algorithm terminates with the result that the solutions of the constraint satisfaction problem consist of tuples with $A = t$ and $C = f$. To also determine the possible values for B , we have to join this result table with the previous constraint we computed for A, B, C :

A	C	A	B	C
t	f	t	f	t
		t	t	f
		f	t	t

This means that there exists exactly one solution to the problem: $A = t, B = t, C = f$.

Exercise 6 :

Consider the cryptarithmic puzzle

$$\begin{array}{r}
 \text{T} \quad \text{W} \quad \text{O} \\
 + \quad \text{T} \quad \text{W} \quad \text{O} \\
 \hline
 \text{F} \quad \text{O} \quad \text{U} \quad \text{R}
 \end{array}$$

Each letter in a cryptarithmic problem represents a different digit; note that F cannot be 0.

- Construct a constraint network representation for the puzzle. *Hint:* For each of the four columns in the table representation above we have a constraint. E.g. for the first column we have the constraint

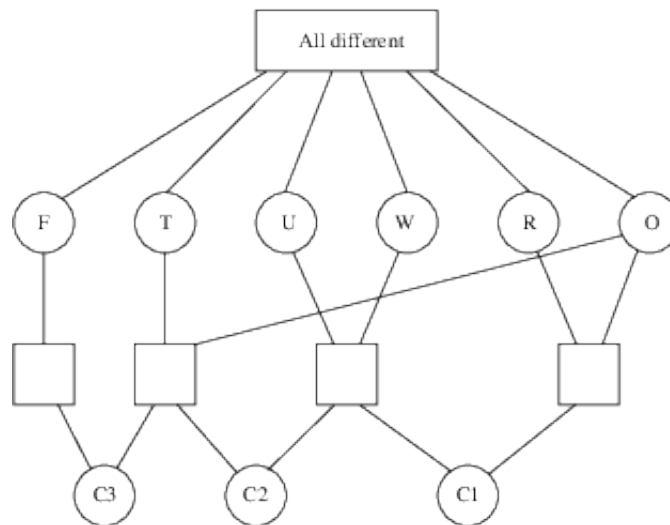
$$O + O = R + 10 \cdot C_1,$$

where C_1 is an auxiliary variable representing what is carried over in the 10 column. Use similar auxiliary variables, say C_2 and C_3 , for encoding what is carried over to the 100 and 1000 column, respectively.

- Perform a BacktrackingWithInference to find a solution to the puzzle. *Hint:* The ordering of the variables has a big impact on the solution size, so a good strategy would be to choose the variable with the smallest number of possible values. E.g. consider starting with $C_3 = 1$.

Solution:

- Here is a constraint network for the cryptarithmic problem



The constraint connecting R , O , and C_1 encodes that $O + O = R + 10 \cdot C_1$. Similar for the other constraints connected to the auxiliary variables.

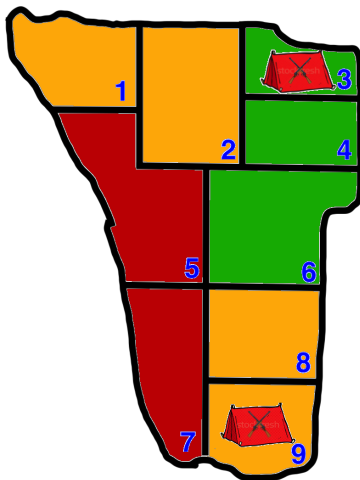
- There are many possible solutions to the problem. One possibility is to
 1. Set $C_3 = 1$.
 2. Set $F = 1$ (the only possible value after applying consistency).
 3. Set (arbitrarily) $C_2 = 0$.
 4. Set (arbitrarily) $C_1 = 0$.
 5. By arc consistency, O must be an even number (it is equal to $T + T$) and less than 5 since $C_1 = 0$. Set (arbitrarily) $O = 4$.

6. By arc consistency, R now only has one value, namely 8.
7. By arc consistency, T only has one value, namely 7.
8. By arc consistency, U must be an even number less than 9. Set $U = 6$.
9. By arc consistency, set $W = 3$.

Exercise 7 :

(Extra exercise for modelling CSPs)

You are a ranger and own a piece of land, depicted below. You want to make sure that all of your animals survive. This is only possible if no animals that kill each other live in the same or neighboring regions. You own rhinos (R), elephants (E), cheetahs (C), lions (L), jackals (J), springbok (S), zebras (Z), hippos (H) and giraffes (G). Your piece of land contains three different climatic regions: bush (green), desert (red), savannah (orange). Some animals need specific climatic environments. Unfortunately there are also two camps of poachers on the land, who are selling rhino horn and elephant tusk. Poachers treacherously shoot animals in their own and neighboring regions.



- Poachers kill rhinos and elephants
- Jackals eat springbok
- Cheetahs eat springbok and zebras
- Lions eat springbok, zebras and giraffes, and sometimes even kill elephants

- Jackals and cheetahs can live in the desert and in the savannah
- Elephants can only live in the bush
- Lions and zebras can live in the bush and in the savannah
- Springbok and hippos only live in the savannah regions

Formulate the problem as a constraint network. Use the set of variables $V = \{v_1, \dots, v_9\}$, where variable v_i models which animal can live in region i .

- (i) Specify the domain D_{v_i} of each variable $v_i \in V$. That is, for each region type give the set of animals that can live in that region, and state which variables have this domain.
- (ii) Specify the binary constraints $C_{v_i v_j} \subseteq D_{v_i} \times D_{v_j}$ for all pairs of variables. That is, for each combination of regions give the set of animal-pairs that the region pair can inhabit, and state which variable pairs have this constraint.
- (iii) Specify the unary constraints $C_{v_i} \subseteq D_{v_i}$ for all relevant variables. That is, for each region affected by the poachers, give the set of animals that can inhabit that region.

Hint: You can specifying constraints with those pairs that are not contained:

$$C_{v_i, v_j} = D_{v_i} \times D_{v_j} \setminus \{(d_{v_i, k}, d_{v_j, l}), \dots\}$$

Solution: Solution:

- (i)
 - $D_S = D_{v_1} = D_{v_2} = D_{v_8} = D_{v_9} = \{R, J, C, S, Z, L, G, H\}$
 - $D_B = D_{v_3} = D_{v_4} = D_{v_6} = \{R, E, Z, L, G\}$
 - $D_D = D_{v_5} = D_{v_7} = \{R, J, C, G\}$
- (ii)
 - $C_{sav, sav} = C_{v_1 v_2} = C_{v_8 v_9}$
 $= D_S \times D_S \setminus \{(S, J), (J, S), (C, S), (S, C), (C, Z), (Z, C),$
 $(L, S), (L, Z), (L, G), (G, L), (Z, L), (S, L)\}$
 - $C_{des, des} = C_{v_5 v_7}$
 $= D_D \times D_D$
 - $C_{bush, bush} = C_{v_3 v_4} = C_{v_4 v_6}$
 $= D_B \times D_B \setminus \{(L, Z), (L, G), (L, E), (Z, L), (G, L), (E, L)\}$
 - $C_{sav, des} = C_{v_1 v_5} = C_{v_2 v_5} = C_{v_8 v_7} = C_{v_9 v_7}$
 $= D_S \times D_D \setminus \{(S, J), (S, C), (Z, C), (L, G)\}$
 - $C_{sav, bush} = C_{v_2 v_3} = C_{v_2 v_4} = C_{v_8 v_6} = C_{v_2 v_6}$
 $= D_S \times D_B \setminus \{(C, Z), (L, Z), (L, G), (L, E), (S, L), (Z, L), (G, L)\}$
 - $C_{des, bush} = C_{v_5 v_6}$
 $= D_D \times D_B \setminus \{(C, Z), (G, L)\}$
- (iii)
 - $D_2 = D_8 = D_9 = D_S \setminus \{R\} = \{J, C, S, Z, L, G, H\}$
 - $D_4 = D_3 = D_B \setminus \{R, E\} = \{Z, L, G\}$
 - $D_7 = D_D \setminus \{R\} = \{J, C, G\}$

Exercise 8 :

(Extra exercise to practice arc consistency (a bit long))

On your piece of land you additionally own a lodge for tourists and a cat sanctuary. You want to show your tourists all your nice animals and everything the region has to offer. You have 5 different time slots: 1, 2, 3, 4, 5, and want to fit all the tasks (see below) in, such that all of the constraints (see below) are satisfied.

Tasks:

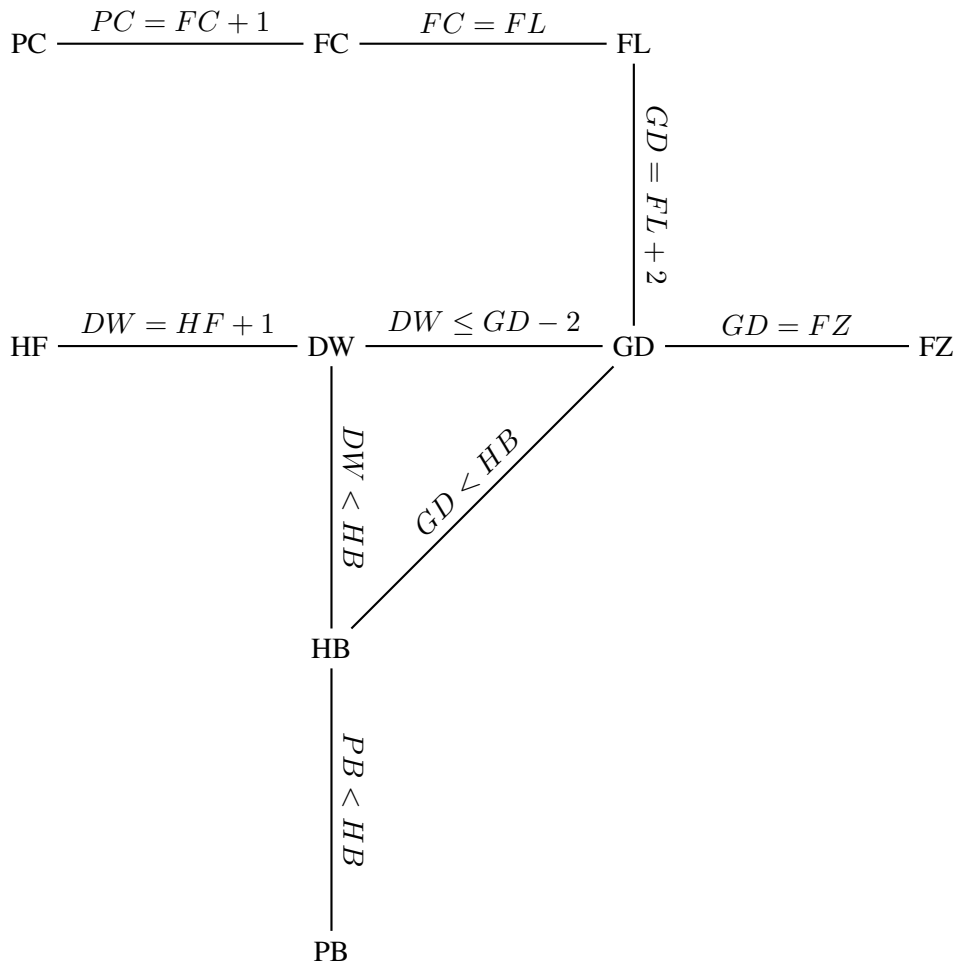
- (GD) go for game drive with tourists
- (DW) go to the desert with tourists
- (HF) make helicopter flight with tourists
- (HB) have braai with tourists
- (PB) prepare braai

- (PC) pet cheetahs
- (FL) feed lions
- (FC) feed cheetahs
- (FZ) feed zebra

Constraints:

- Cheetahs need to be fed immediately before you can pet them ($FC = PC - 1$)
 - Lions and cheetahs have to be fed at the same time, otherwise they will get jealous ($FC = FL$)
 - The desert walk has to be (at least) two time slots before the game drive, otherwise your tourists will get to exhausted ($DW \leq GD - 2$)
 - In order to have braai with your tourists you first have to exhaust them on the game drive and on the desert walk, so they can enjoy their beer ($GD < HB$; $DW < HB$)
 - You have to prepare braai before having the braai with your tourists ($PB < HB$)
 - You can only find lions on the game drive if they have been fed two time slots earlier ($GD = FL + 2$)
 - In order to give your tourists an impression of the area first, the heli flight is done just before the desert walk ($HF = DW - 1$)
 - In order to save time you want to feed the zebra on the game drive ($FZ = GD$)
 - Variables: $V = \{DW, FC, FL, FZ, GD, HB, HF, PC, PB\}$
 - Domains: For all $v \in V$: $D_v = \{1, 2, 3, 4, 5\}$
- (i) Draw the constraint graph of γ . Label each edge with the corresponding constraint.
- (ii) Run Generalized Arc consistency. Initialize To-do-arcs with an alphabetical list. For each step: Give the selected pair, the updated domain, and the pairs that are added to To-do-arcs (give all pairs that would be added and mark those that are not already contained). You do not have to give the updated To-do-arcs list in each iteration.

Solution: <i>Solution:</i>



(i)

- (ii) $M = \{(DW, GD), (DW, HB), (DW, HF), (FC, FL), (FC, PC), (FL, FC), (FL, GD), (FZ, GD), (GD, DW), (GD, FL), (GD, FZ), (GD, HB), (HB, DW), (HB, GD), (HB, PB), (HF, DW), (PB, HB), (PC, FC)\}$

- (1) **selected:** (DW, GD) ; $D_{DW} = \{1, 2, 3\}$; **add:** $(HB, DW), (HF, DW)$
- (2) **selected:** (DW, HB) ; no changes
- (3) **selected:** (DW, HF) ; $D_{DW} = \{2, 3\}$; **add:** $(GD, DW), (HB, DW)$
- (4) **selected:** (FC, FL) ; no changes
- (5) **selected:** (FC, PC) ; $D_{FC} = \{1, 2, 3, 4\}$; **add:** (FL, FC)
- (6) **selected:** (FL, FC) ; $D_{FL} = \{1, 2, 3, 4\}$; **add:** (GD, FL)
- (7) **selected:** (FL, GD) ; $D_{FL} = \{1, 2, 3\}$; **add:** (FC, FL)
- (8) **selected:** (FZ, GD) ; no changes
- (9) **selected:** (GD, DW) ; $D_{GD} = \{4, 5\}$; **add:** (FL, GD) , (FZ, GD) , (HB, GD)
- (10) **selected:** (GD, FL) ; no changes

- (11) **selected:** (GD, FZ) ; no changes
- (12) **selected:** (GD, HB) ; $D_{GD} = \{4\}$; **add:** (DW, GD) , (FL, GD) , (FZ, GD)
- (13) **selected:** (HB, DW) ; $D_{HB} = \{3, 4, 5\}$; **add:** (GD, HB) , (PB, HB)
- (14) **selected:** (HB, GD) ; $D_{HB} = \{5\}$; **add:** (DW, HB) , (PB, HB)
- (15) **selected:** (HB, PB) ; no changes
- (16) **selected:** (HF, DW) ; $D_{HF} = \{1, 2\}$; no adds
- (17) **selected:** (PB, HB) ; $D_{PB} = \{1, 2, 3, 4\}$; no adds
- (18) **selected:** (PC, FC) ; $D_{PC} = \{2, 3, 4, 5\}$; no adds
- (19) **selected:** (FC, FL) ; $D_{FC} = \{1, 2, 3\}$; **add:** (PC, FC)
- (20) **selected:** (FL, GD) ; $D_{FL} = \{2\}$; **add:** (FC, FL)
- (21) **selected:** (FZ, GD) ; $D_{FZ} = \{4\}$; no adds
- (22) **selected:** (DW, GD) ; $D_{DW} = \{2\}$; **add:** (HB, DW) , (HF, DW)
- (23) **selected:** (GD, HB) ; no changes
- (24) **selected:** (DW, HB) ; no changes
- (25) **selected:** (PC, FC) ; $D_{PC} = \{2, 3, 4\}$; no adds
- (26) **selected:** (FC, FL) ; $D_{FC} = \{2\}$; **add:** (PC, FC)
- (27) **selected:** (HB, DW) ; no changes
- (28) **selected:** (HF, DW) ; $D_{HF} = \{1\}$; no adds
- (29) **selected:** (PC, FC) ; $D_{PC} = \{3\}$; no adds

M empty; return modified γ :

$$D_{DW} = \{2\}$$

$$D_{FC} = \{2\}$$

$$D_{FL} = \{2\}$$

$$D_{FZ} = \{4\}$$

$$D_{GD} = \{4\}$$

$$D_{HB} = \{5\}$$

$$D_{HF} = \{1\}$$

$$D_{PB} = \{1, 2, 3, 4\}$$

$$D_{PC} = \{3\}$$